

MANAGEMENT ALGORITHM FOR
THROMBOCYTOSIS AND ERYTHROCYTOSIS
IN PRIMARY CARE

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MANAGEMENT ALGORITHM FOR THROMBOCYTOSIS IN PRIMARY CARE

Common causes of secondary thrombocytosis

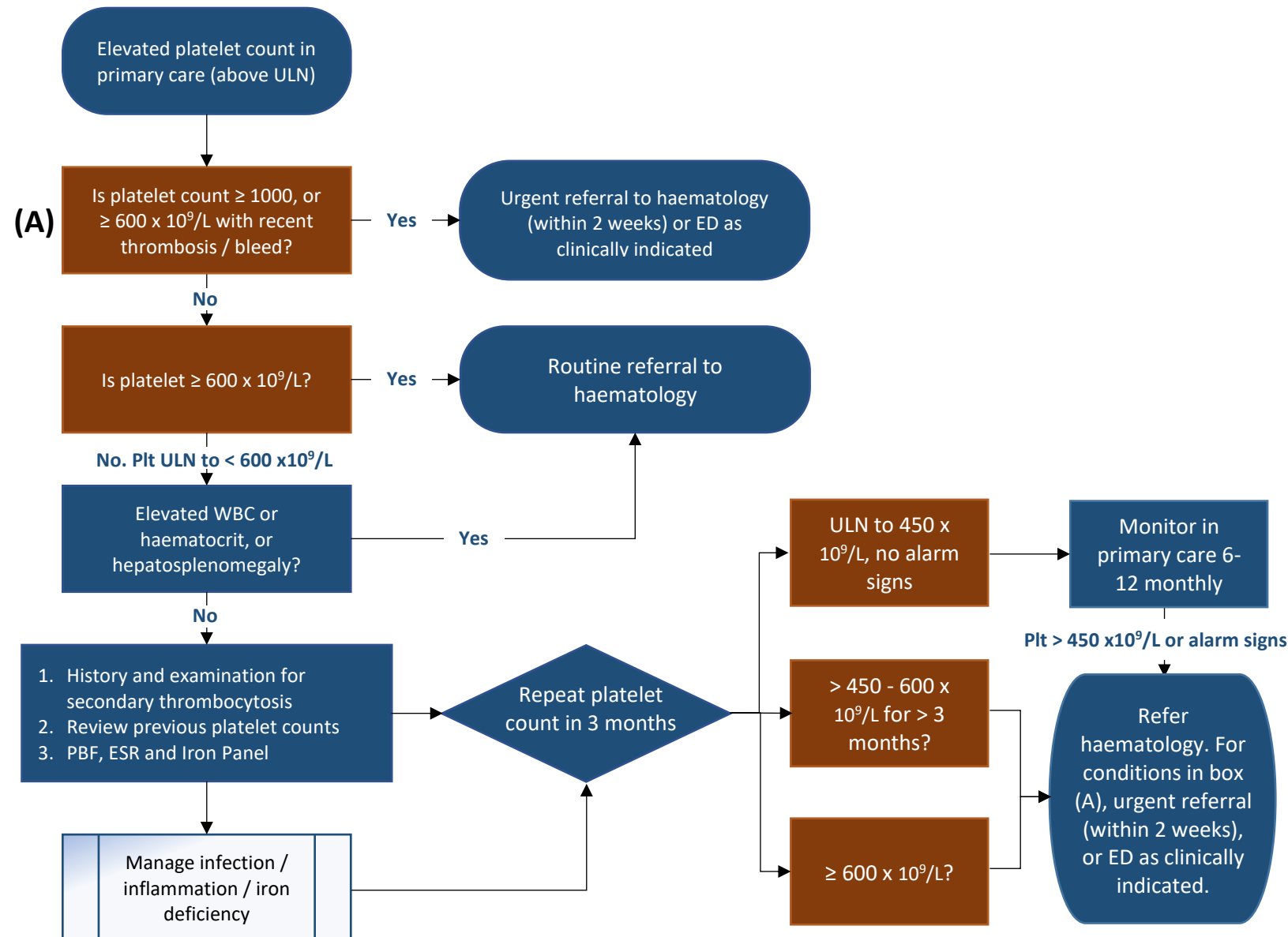
1. Iron deficiency
2. Inflammation
3. Infection
4. Recent blood loss/surgery
5. Prior splenectomy
6. Borderline high normal variant

Common causes of primary thrombocytosis

1. Essential thrombocytosis
2. Chronic myeloid leukaemia

Alarm signs requiring haematology referral to exclude myeloproliferative disease

1. $\geq 600 \times 10^9/L$ (urgent if ≥ 1000 or thrombosis/bleeding)
2. $> 450 - 600 \times 10^9/L$ for > 3 months
3. Hepatosplenomegaly
4. Significantly elevated WBC and/or haematocrit
5. Abnormal peripheral blood film (PBF)



MANAGEMENT ALGORITHM FOR ERYTHROCYTOSIS IN PRIMARY CARE

*Exclude relative polycythemia, i.e. dehydration, diuretic use. Manage accordingly then follow the algorithm if not resolved

Common causes of secondary erythrocytosis

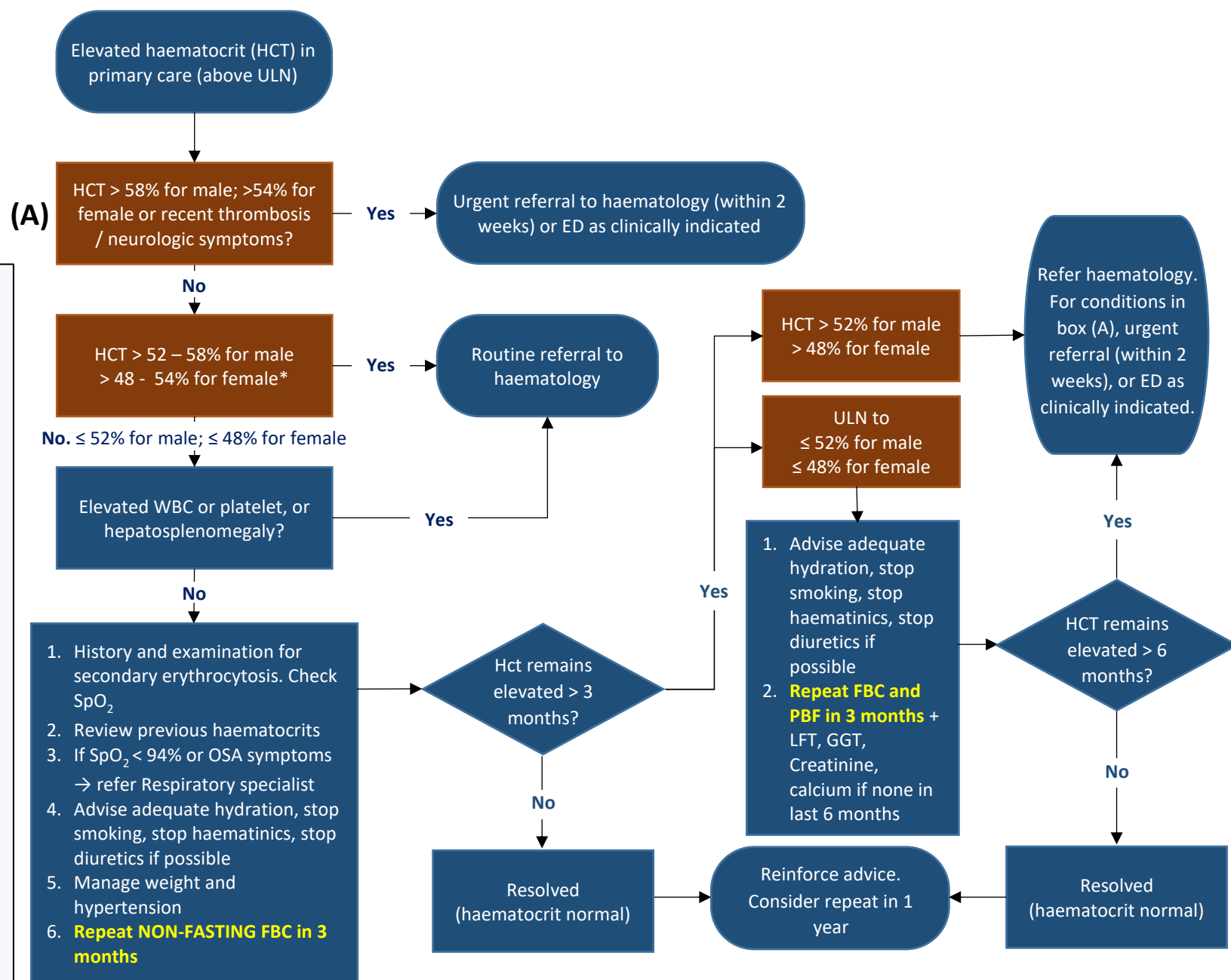
1. Dehydration/ diuretics/ fasting
2. Smoking
3. Gaisbock syndrome (obesity, hypertension)
4. Obstructive sleep apnoea
5. Liver and kidney cysts
6. Borderline high normal variant
7. Chronic hypoxic states (e.g. COPD, right to left cardiac shunt)

Common causes of primary erythrocytosis

1. Polycythemia rubra vera
2. Idiopathic erythrocytosis

Alarm signs requiring haematology referral to exclude polycythemia rubra vera

1. Recent thrombosis or neurologic symptoms due to elevated HCT (i.e. headache, dizziness, blurring of vision)
2. > 52% for males, > 48% for females
3. Hepatosplenomegaly
4. Significantly elevated WBC and/or platelets
5. Abnormal peripheral blood film (PBF)



FREQUENTLY ASKED QUESTIONS (FAQ)

Thrombocytosis

- 1) The threshold for referral to haematology is if platelet counts exceed $450 \times 10^9/L$ but our laboratory upper limit of normal is set lower at $360 \times 10^9/L$. Is there a need to refer patients with platelet counts between $360 \times 10^9/L$ and $450 \times 10^9/L$ to haematology?

Answer: Majority of patients with platelet counts below $450 \times 10^9/L$ and not increasing over years are likely to have a benign cause. For essential thrombocytosis, the platelet count criteria is sustained over $> 450 \times 10^9/L$. For other myeloproliferative diseases which may present with thrombocytosis, they will most likely show an increasing trend over several months to years.

- 2) Hematologic malignancies may present with constitutional symptoms of loss of weight, night sweats, pruritus, flushing or erythromelalgia. Should we include these alarm symptoms in the workflow as well?

Answer: Majority do not have symptoms at presentation. By the time they have symptoms, there will usually be other abnormalities mentioned in alarms signs. By all means, continue to ask about these symptoms as holistic assessment.

Erythrocytosis

- 1) What is the rationale for doing LFT and calcium for erythrocytosis work up?

Answer: It is to screen for rare causes like erythropoietin producing tumours such as hepatocellular carcinoma and parathyroid adenoma / carcinoma.

- 2) For patients who are referred to ENT / Respiratory Medicine for OSA work up or suspected chronic hypoxic states, do we still proceed with the rest of the workflow?

Answer: Yes. Patients should still be followed up and haematocrit monitored because specialists may not agree that erythrocytosis is the result of the suspected problem e.g. Obstructive Sleep Apnoea and/or the patient may have another concurrent cause of erythrocytosis.

- 3) If we find persistent erythrocytosis on reviewing previous haematocrits, which part of the algorithm should we start on?

Answer: If patient is found to have persistent erythrocytosis for > 3 months or > 6 months, the patient would already be further down the algorithm at the "HCT remains elevated > 3 months or > 6 months" diamonds and hence should be managed from that point onwards.

- 4) What are the actions to take when patients are seen at the 3-month and 6-month mark in the flowchart?

Answer: It is important to ensure that the haematocrit is taken on a non-fasting sample because dehydration from the fasting state together with use of certain oral hypoglycaemic agents and/or diuretic medications can lead to haemoconcentration and spuriously elevated haematocrit readings. Once the patients are established to have an elevated haematocrit but not at the level / no alarm signs for immediate haematology referral, management should include advising the patient to get adequate hydration, stop smoking, reviewing medications with a view of stop haematinics and diuretics if possible, and managing hypertension and obesity if any. Such measures may be effective in reducing haematocrit over a 3-6 month period. Some patients with Gaisbock syndrome (association of metabolic syndrome, obesity and hypertension with polycythaemia) may take more than 6 months for their haematocrit to decline, if ever. Nonetheless, if haematocrit is persistently elevated > 6 months, a referral to haematology is warranted to exclude primary polycythaemia rubra vera or idiopathic erythrocytosis before attributing it to benign causes.

ABBREVIATIONS

ED:	Emergency department
ET:	Essential thrombocytosis
HCT:	Haematocrit
LFT:	Liver function test
OSA:	Obstructive sleep apnoea
PBF:	Peripheral blood film
RP:	Renal panel
ULN:	Upper limit of normal
WBC:	White blood cell

REFERENCES

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